

FARMARE — automated on-chain yield, engineered for the worst regime

A rules-based DeFi strategy on Ethereum L2s combining a diversified stablecoin core with a hedged liquidity-provision engine. Optimized with a maximin criterion: every design decision was selected for its worst-case market regime, not its best.

+7.5% APY, worst tested regime (real semester, ETH -44%)	+13–22% engine APY, every calendar year 2023–2026	≤ -1% portfolio max drawdown, historical window	7/7 stress scenarios pass in production code
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Simulated results net of modeled costs. Live paper trading since 8 July 2026 — every decision git-committed and auditable.

How it works

CORE — 80%, stablecoin yield. Rotates across whitelisted blue-chip venues (Aave, Morpho, Pendle, Fluid...), split over two protocols (max ~50% each). Locks **fixed rates** via Pendle PT when fixed beats floating by ≥1pp — backtest: +11.5% vs +5.6% floating over the same window.

ENGINE — 20%, hedged market-making. Concentrated ETH/USDC liquidity on Uniswap v3 with a volatility-adaptive range and a conditional short-perp hedge: the LP becomes a fee-carry trade, not a directional bet. Switches to delta-neutral funding capture when funding clearly pays more.

Outcome range by regime (backtested, net of costs)

The strategy is selected on its worst regime — but most regimes are not the worst. Full validated range:

Market regime	Portfolio (Prudent 80/20)	Engine sleeve
Bear — real semester, ETH -44%	+7.5% APY (the floor)	+17.6% APY
Sideways (synthetic, mean of paths)	+16.3% APY	+28.5% APY
Bull (synthetic, +60% drift)	+10.0% APY	+8.1% APY
Engine, full period 2023–2026	—	+22.5% APY, every year positive
Decade 2016–2026, guarded engine	positive every year , worst +3.7%	engine active only when viable

Evidence

Test	Data	Result
Real bear semester (ETH -44%)	Real pool volumes, prices, funding	Hedged LP +26% vs hold , -5% max drawdown
Three regimes (bear/flat/bull)	Real + synthetic paths	Prudent 80/20 never worse than -2pp vs any alternative
Three chains, same strategy	Base · Solana · BSC, real data	Hedge thesis confirmed on all three
Long history 2023–2026	Binance prices, HL funding, calibrated volumes	Positive every year ; +22.5% full period

Fee model vs on-chain reality	Live RPC measurement, daily	Model conservative (0.53 measured vs 0.60 assumed)
Ideas rejected by the data	Maximin across regimes	6 (incl. trend filters and regime switching)

Risk framework

Hourly automated guards with immediate alerts and ≤ 6 h automatic reaction: underlying **depeg** ($< \$0.99 \rightarrow$ emergency exit), **TVL collapse** (-30% vs 7-day high), protocol **whitelist**, data sanity, fee-model drift, hedge margin stress. The diversification invariant (two core slots never share a protocol) is enforced in code and covered by the stress suite.

Quantified worst case	Portfolio impact
Single-candle -35% ETH gap (March-2020 grade)	$\approx -2.4\%$
Stable depeg while holding (detection ≤ 1 h)	Exit cost $\approx 0.23\%$ of the slot + drift
Full data-API outage	0 — run aborts, positions static, catch-up on resume

Acceptance gates before real capital

Gate	Threshold	Status
Stress suite	7/7 green on every release	passing
Fee-model drift	Within $\pm 25\%$ of assumption	passing (-12%)
Forward test	≥ 4 weeks, APY within model error	running
Phase-2 vault (ERC-4626)	Fork-tested + external review	scheduled
Legal structuring	MiCA-compliant wrapper for third-party funds	required

What we tell you before you ask. These are simulated results on live market data — no client funds are deployed. Fee capture uses a calibrated model (verified daily on-chain, currently conservative). Long-history volumes are a calibrated proxy with a measured ± 2 pp error bar. Accepting third-party funds with fees is a regulated activity: nothing here is an offer, and no deposit will be accepted before the gates above are green and legal structuring is complete.

Operations & auditability

Runs 4x/day plus hourly safety checks in cloud CI, independent of any personal machine. Every decision — each rotation, hedge adjustment, emergency exit — is journaled and **committed to git**: the track record is timestamped and tamper-evident. Wealth-manager-grade reporting on every capital movement (amount, source, destination, reason, cost). Share-based fund accounting (NAV, high-water mark, entry / management / performance fees) is built and tested, dormant until authorized.

FARMARE — private research project. Full methodology, limitations and reproduction instructions: DUE_DILIGENCE.md in the repository; live dashboard regenerated on every run. This document is informational only and is not investment advice or an offer of any financial product.